UFCFK1-15-0 - Fundamentals of data science 25feb_1

Take home assessment 40% (PSA)

Documentation 3

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Section : A

GIT : https://github.com/999-ztj/FODS_assignments

Assignment 3 - File Handling & OOP

Question 1

Program Description:

This program reads the file "records.csv" and displays its contents.The fields are: student_name, roll_no, program, year, and group.

Screenshot:

```
=== Student Records ===

student_name,roll_no,program,year,group
--------------------------------------------------
Name         : Ivan Tamrakar
Roll No      : TBC001
Program      : BSc AI
Year         : 2
Group        : A
--------------------------------------------------
Name         : Aayatish Gautam
Roll No      : TBC002
Program      : BSc CS
Year         : 1
Group        : B
--------------------------------------------------
Name         : Jasmine karki
Roll No      : TBC003
Program      : BIT
Year         : 3
Group        : A
--------------------------------------------------
Name         : Jenith Gurung
Roll No      : TBC004
Program      : BSc AI
Year         : 2
Group        : C
--------------------------------------------------
Name         : Saranshika Kc
Roll No      : TBC005
Program      : BCA
Year         : 1
Group        : A
--------------------------------------------------
```

Question 2

Program Description:

This program takes student details as input from the user and appends them to the existing file "records.csv".Fields: student_name, roll_no, program, year, and group.

Screenshot:

```
=== Add New Student Record ===

Enter student name: Jenith.jr Gurung
Enter roll number: TBC017
Enter program: Bsc AI
Enter year: 1
Enter group: A

Record added successfully!
Added: Jenith.jr Gurung,TBC017,Bsc AI,1,A
```

	student_name	roll_no	program	year	group
1	Ivan Tamrakar	TBC001	BSc AI	2	A
2	Aayatish Gautam	TBC002	BSc CS	1	B
3	Jasmine karki	TBC003	BIT	3	A
4	Jenith Gurung	TBC004	BSc AI	2	C
5	Saranshika Kc	TBC005	BCA	1	A
6	Jenith.jr Gurung	TBC017	Bsc AI	1	A

Question 3

Program Description:

This program accepts two integers from the user, performs addition, subtraction, multiplication, and division, and saves results to a file with the current date and time. It keeps running until the user exits.

Screenshot:

```
=== Calculator with File Saving ===
Type 'exit' when asked for first number to quit.

Enter first number (or 'exit' to quit): 17
Enter second number: 19
Results saved at 2026-04-17 19:44:13

Enter first number (or 'exit' to quit): exit

=== Saved Results ===
Date/Time   : 2026-04-17 19:44:13
Numbers     : 17 and 19
Addition    : 36
Subtraction: -2
Multiply    : 323
Division    : 0.8947368421052632
----------------------------------------
```

q3_operations_file.py results.csv

Edit in Data Wrangler

```
C1
Date/Time   : 2026-04-17 21:07:40
Numbers     : 17 and 19
Addition    : 36
Subtraction: -2
Multiply    : 323
Division    : 0.8947368421052632
----------------------------------------
```

Question 4

Program Description:

This program reads content from one file and copies it into another file.The user enters both file names. It uses try/except to handle errors such as an input file not found or an output file already existing.

Screenshot:

```
=== File Copy Program ===

Enter the name of the input file: data.txt
Enter the name of the output file: copy.txt

Error: Input file does not exist.
[Errno 2] No such file or directory: 'data.txt'

Process finished with exit code 0
```

```
=== File Copy Program ===

Enter the name of the input file: test.txt
Enter the name of the output file: copy.txt

File copied successfully from 'test.txt' to 'copy.txt'.

Process finished with exit code 0
```

Question 5

Program Description:

This program creates a class called Learner with attributes: roll_no, full_name, address, enrollment_year, program, and group.It takes input for all attributes and displays the details.

Screenshot:

```
Enter Learner Details:

Roll No          : 17
Full Name        : Jenith Gurung
Address          : Koteshwor
Enrollment Year  : 2025
Program          : Bsc AI
Group            : A

=== Learner Details ===
Roll No          : 17
Full Name        : Jenith Gurung
Address          : Koteshwor
Enrollment Year  : 2025
Program          : Bsc AI
Group            : A

Process finished with exit code 0
```

Question 6

Program Description:

This program creates a Dish class with attributes: dish_id, dish_name,ingredients, and instructions. A Cookbook class manages a collection of Dish objects with options to add and display dishes.

Screenshot:

```
=== Add Dishes to Cookbook ===

Do you want to add a dish? (yes/no): yes
  Dish ID      : 1719
  Dish Name    : Bibimbap
  Ingredients  : Rice,gochujang,veggies
  Instructions : mix all ingredients together
Dish 'Bibimbap' added to cookbook.

Do you want to add a dish? (yes/no): yes
  Dish ID      : 1917
  Dish Name    : pasta
  Ingredients  : pasta,veggies
  Instructions : boil it
Dish 'pasta' added to cookbook.

Do you want to add a dish? (yes/no): no

=== All Dishes in Cookbook ===
  Dish ID      : 1719
  Dish Name    : Bibimbap
  Ingredients  : Rice,gochujang,veggies
  Instructions : mix all ingredients together

  Dish ID      : 1917
  Dish Name    : pasta
  Ingredients  : pasta,veggies
  Instructions : boil it
```

Question 7

Program Description:

This program creates a Staff class with attributes: emp_id, full_name, address, phone_number, marital_status, dependents, and salary.It allows adding multiple staff members, saves them to staff.csv, and displays the list of staff. Uses try/except for error handling.

Screenshot:

```
=== Staff Management System ===

1. Add Staff Member
2. View All Staff
3. Exit
Enter choice: 1

Employee ID    : 1719
Full Name      : Jenith Gurung
Address        : KTM
Phone Number   : 9765828069
Marital Status : Married
Dependents     : 2
Salary         : 100000
Staff data saved to staff.csv

1. Add Staff Member
2. View All Staff
3. Exit
Enter choice: 2

=== Staff List ===
------------------------------------------------------------
emp_id, full_name, address, phone_number, marital_status, dependents, salary
1719, Jenith Gurung, KTM, 9765828069, Married, 2, 100000
------------------------------------------------------------

1. Add Staff Member
2. View All Staff
3. Exit
Enter choice: 3
Exiting program.

Process finished with exit code 0
```

Question 8

Program Description:

This program is a basic Library Management System.It allows users to: borrow a book, return a book, and search for a book.It uses OOP (classes), file handling for storing book data, and try/except for exception handling.

Screenshot:

```
=== Library Management System ===

1. Search for a Book
2. Borrow a Book
3. Return a Book
4. View All Books
5. Exit

Enter your choice: 4

=== All Books ===
------------------------------------------------------------
ID: B001 | Introduction to Python | John Smith | Available
ID: B002 | Data Science Basics | Surav joshi | Available
ID: B003 | Machine Learning | Jenith Gurung | Available
ID: B004 | Database Systems | Sara kc | Available
------------------------------------------------------------
```

```
1. Search for a Book
2. Borrow a Book
3. Return a Book
4. View All Books
5. Exit

Enter your choice: 1
Enter book title or author to search: Machine Learning

Search results for 'Machine Learning':
--------------------------------------------------
ID: B003 | Title: Machine Learning | Author: Jenith Gurung | Available
```

```
1. Search for a Book
2. Borrow a Book
3. Return a Book
4. View All Books
5. Exit

Enter your choice: 2

=== All Books ===
------------------------------------------------------------
ID: B001 | Introduction to Python | John Smith | Available
ID: B002 | Data Science Basics | Surav joshi | Available
ID: B003 | Machine Learning | Jenith Gurung | Available
ID: B004 | Database Systems | Sara kc | Available
------------------------------------------------------------
Enter Book ID to borrow: B004
You have borrowed 'Database Systems'. Please return it on time.
```

```
1. Search for a Book
2. Borrow a Book
3. Return a Book
4. View All Books
5. Exit

Enter your choice: 3

=== All Books ===
------------------------------------------------------------
ID: B001 | Introduction to Python | John Smith | Available
ID: B002 | Data Science Basics | Surav joshi | Available
ID: B003 | Machine Learning | Jenith Gurung | Available
ID: B004 | Database Systems | Sara kc | Borrowed
------------------------------------------------------------
Enter Book ID to return: B004
Thank you for returning 'Database Systems'.
```

```
1. Search for a Book
2. Borrow a Book
3. Return a Book
4. View All Books
5. Exit

Enter your choice: 5
Thank you for using the Library System. Goodbye!

Process finished with exit code 0
```